

CONTACT INFORMATION	Center for the Fundamental Laws of Nature Harvard University	<i>e-mail:</i> grant.remmen@gmail.com <i>web:</i> grantremmen.com
POSITIONS	Harvard University , Research Associate New York University , James Arthur Postdoctoral Fellow University of California, Santa Barbara , Fundamental Physics Fellow Kavli Institute for Theoretical Physics , Postdoctoral Scholar University of California, Berkeley , Miller Research Fellow Harvard University Society of Fellows , Junior Fellow (declined)	2026–present 2023–2026 2020–2023 2017–2020 2017
EDUCATION	California Institute of Technology Ph.D., Physics M.S., Physics Hertz Fellow and NSF Graduate Research Fellow University of Minnesota, College of Science & Engineering B.S., Physics, <i>summa cum laude</i> , High Distinction, 4.0 GPA B.S., Astrophysics, <i>summa cum laude</i> , High Distinction, 4.0 GPA B.S., Mathematics, <i>summa cum laude</i> , High Distinction, 4.0 GPA	2012–2017 June 2017 June 2015 2008–2012 May 2012 May 2012 May 2012
SELECTED HONORS & AWARDS	Postdoctoral Research and Professional Development Support Grant , NYU Appointed as Hertz Fellowship Interviewer Sakurai Dissertation Award in Theoretical Particle Physics American Physical Society award citation: “For his contributions to understanding the structure and self-consistency of gravity and effective field theories using ideas from quantum field theory and holography.” Stemple Memorial Prize in Physics , Caltech United States Delegate to the 66th Lindau Nobel Laureate Meeting Hertz Fellow NSF Graduate Research Fellow , National Science Foundation Goldwater Scholar Chambliss Astronomy Achievement Student Award , American Astronomical Society Dean’s Summer International Student Scholarship , University College London National Merit Scholar Byrd Honors Scholar United States Presidential Scholar , White House Commission on Presidential Scholars & U.S. Dept. of Education	2025 2018–present 2018 2016 2016 2012–2017 2012–2017 2010–2012 2011 2011 2008–2012 2008–2011 2008
SCIENTIFIC & HONORARY AFFILIATIONS	American Physical Society American Astronomical Society International Society on General Relativity and Gravitation Sigma Xi, The Scientific Research Honor Society Sigma Pi Sigma, National Physics Honor Society	

PRESS	Phys.Org & NYU News <i>Physicists Turn to the Universe's "Piano Notes" to Detect Hidden Particles</i>	July 2026
	Physics Magazine, APS <i>Can String Theory Be Explained with No Strings Attached?</i>	June 2026
	Popular Mechanics <i>Scientists Think They Just Accidentally Proved String Theory</i>	May 2026
	Phys.Org <i>String Theory Is Uniquely Derived from Basic Assumptions About the Universe, Physicists Show</i>	May 2026
	Caltech News & NYU News <i>String Theory Emerges from "Almost Nothing"</i>	May 2026
	Science News <i>To Get String Theory, You Need Only Four Physics Assumptions</i>	May 2026
	Quanta Magazine <i>Are Strings Still Our Best Hope for a Theory of Everything?</i>	March 2026
	Physics World, IOP, Editor's Choice <i>String Theory May Be Inevitable as a Unified Theory of Physics, Calculations Suggest</i>	January 2025
	NYU News <i>Physicists 'Bootstrap' Validity of String Theory</i>	December 2024
	Phys.org <i>Theoretical Study Shows That Kerr Black Holes Could Amplify New Physics</i>	September 2023
	Physics Magazine, APS <i>New Physics Magnified in Spinning Black Holes</i>	August 2023
	The Current, UCSB <i>Quantum Zeta Epiphany</i>	January 2022
	Physics Magazine, APS <i>A Physical Match for the Riemann Zeta Function</i>	December 2021
	Quanta Magazine <i>Black Hole Paradoxes Reveal a Fundamental Link Between Energy and Order</i>	May 2020
PUBLICATIONS	76. Grant N. Remmen <i>Zeta Functions and the Superstring</i>	under review, Phys. Rev. Lett. arXiv:2606.09977
	75. Gary T. Horowitz, Maciej Kolanowski, Grant N. Remmen , Jorge E. Santos <i>Axions Create Singularities on Extremal Horizons</i>	under review, Phys. Rev. Lett. arXiv:2605.30411
	74. Clifford Cheung, Jaehoon Jeong, Pyungwon Ko, Alex Pomarol, Grant N. Remmen , Francesco Sciotti <i>Multipositivity Constrains the Chiral Lagrangian</i>	under review, Phys. Rev. Lett. arXiv:2605.21582
	73. Francesco Calisto, Clifford Cheung, Grant N. Remmen , Francesco Sciotti, Michele Tarquini <i>The Equivalence Principle at High Energies Completes the Spectrum</i>	under review, Phys. Rev. Lett. arXiv:2605.20319
	72. Francesco Calisto, Clifford Cheung, Grant N. Remmen , Francesco Sciotti, Michele Tarquini <i>On the Inverse Problem in Effective Field Theory</i>	accepted, Phys. Rev. Lett. arXiv:2604.15423
	71. Ivano Basile, Grant N. Remmen , Georgina Staudt <i>Higher-Spin and Higher-Point Constraints on Stringy Amplitudes</i>	Phys. Rev. D 114 (2026) 026001 arXiv:2603.04485
	70. Francesco Calisto, Clifford Cheung, Grant N. Remmen , Francesco Sciotti, Michele Tarquini <i>Completeness from Gravitational Scattering</i>	Phys. Rev. D 113 (2026) 106008 arXiv:2512.11955
	69. Clifford Cheung, Grant N. Remmen , Francesco Sciotti, Michele Tarquini <i>Strings from Almost Nothing</i>	Phys. Rev. Lett. 136 (2026) 251601 Editors' Suggestion arXiv:2508.09246
	68. Clifford Cheung, Grant N. Remmen <i>Multipositivity Bounds for Scattering Amplitudes</i>	Phys. Rev. D 112 (2025) 016017 arXiv:2505.05553

- PUBLICATIONS, 67. Ning Bao, **Grant N. Remmen** Int. J. Mod. Phys. D **34** (2025) 2544009
 CONTINUED *Black Hole Complementarity and ER/EPR* arXiv:2503.16610
 Honorable Mention, 2025 Awards for Essays on Gravitation, Gravity Research Foundation
66. Nima Arkani-Hamed, Carolina Figueiredo, **Grant N. Remmen** JHEP **4** (2025) 39
Open String Amplitudes: arXiv:2412.20639
Singularities, Asymptotics, and New Representations
65. Avik Banerjee, Achilleas P. Porfyriadis, **Grant N. Remmen** JHEP **4** (2025) 149
Accidental Symmetry Near Extreme Spinning Black Holes arXiv:2412.19880
64. **Grant N. Remmen**, Nicholas L. Rodd Phys. Rev. D **113** (2026) 036027
Positively Identifying Higgs Effective Field Theory or arXiv:2412.07827
Standard Model Effective Field Theory
63. Gauthier Durieux, **Grant N. Remmen**, SciPost Phys. Comm. Rep. **6** (2025)
 Nicholas L. Rodd et al. arXiv:2411.02483
LHC EFT WG Note: Basis for Anomalous Quartic Gauge Couplings
62. Clifford Cheung, Aaron Hillman, **Grant N. Remmen** Phys. Rev. D **111** (2025) 086034
Uniqueness Criteria for the Virasoro-Shapiro Amplitude arXiv:2408.03362
61. Clifford Cheung, Aaron Hillman, **Grant N. Remmen** Phys. Rev. Lett. **133** (2024) 251601
Bootstrap Principle for the Spectrum and Scattering of Strings arXiv:2406.02665
60. Gary T. Horowitz, Maciej Kolanowski, JHEP **5** (2024) 122
Grant N. Remmen, Jorge E. Santos arXiv:2403.00051
Sudden Breakdown of Effective Field Theory Near
Cool Kerr-Newman Black Holes
59. Rafael Aoude, Gilly Elor, **Grant N. Remmen**, Fortschr. Phys. **74** (2026) e70113
 Olcyr Sumensari arXiv:2402.16956
Positivity in Amplitudes and Quantum Entanglement
58. Nima Arkani-Hamed, Clifford Cheung, Phys. Rev. Lett. **132** (2024) 091601
 Carolina Figueiredo, **Grant N. Remmen** arXiv:2312.07652
Multiparticle Factorization and the Rigidity of String Theory
57. Aidan Chatwin-Davies, Pompey Leung, **Grant N. Remmen** Phys. Rev. D **109** (2024) 046003
Holographic Screen Sequestration arXiv:2312.06750
56. Xi Dong, **Grant N. Remmen**, Diandian Wang, JHEP **11** (2023) 207
 Wayne W. Weng, Chih-Hung Wu arXiv:2308.07952
Holographic Entanglement from the UV to the IR
55. Clifford Cheung, **Grant N. Remmen** Phys. Rev. D **108** (2023) 086009
Bespoke Dual Resonance arXiv:2308.03833
54. Gary T. Horowitz, Maciej Kolanowski, Phys. Rev. Lett. **131** (2023) 091402
Grant N. Remmen, Jorge E. Santos Editors' Suggestion
Extremal Kerr Black Holes as Amplifiers of New Physics arXiv:2303.07358
53. Clifford Cheung, **Grant N. Remmen** Phys. Rev. D **108** (2023) 026011
Stringy Dynamics from an Amplitudes Bootstrap arXiv:2302.12263
52. Achilleas P. Porfyriadis, **Grant N. Remmen** JHEP **3** (2023) 125
Charged Dilatonic Spacetimes in String Theory arXiv:2301.08256
51. Clifford Cheung, **Grant N. Remmen** JHEP **1** (2023) 122
Veneziano Variations: How Unique are String Amplitudes? arXiv:2210.12163
50. Marat Freytsis, Soubhik Kumar, **Grant N. Remmen**, Nicholas L. Rodd JHEP **9** (2023) 41
Multifield Positivity Bounds for Inflation arXiv:2210.10791
49. Juan Maldacena, **Grant N. Remmen** JHEP **8** (2022) 152
Accumulation-Point Amplitudes in String Theory arXiv:2207.06426

- PUBLICATIONS, 48. **Grant N. Remmen**, Nicholas L. Rodd JHEP **9** (2022) 30
CONTINUED *Spinning Sum Rules for the Dimension-Six SMEFT* arXiv:2206.13524
47. Yu-tin Huang, **Grant N. Remmen** Phys. Rev. D **106** (2022) L021902
UV-Complete Gravity Amplitudes and the Triple Product arXiv:2203.00696
46. Achilleas P. Porfyriadis, **Grant N. Remmen** JHEP **3** (2022) 107
Large Diffeomorphisms and Accidental Symmetry of the Extremal Horizon arXiv:2112.13853
45. **Grant N. Remmen** Gen. Rel. Grav. **53** (2021) 101
Exploration of a Singular Fluid Spacetime arXiv:2111.08713
44. Nima Arkani-Hamed, Yu-tin Huang, Jin-Yu Liu, **Grant N. Remmen** JHEP **3** (2022) 83
Causality, Unitarity, and the Weak Gravity Conjecture arXiv:2109.13937
43. **Grant N. Remmen** Phys. Rev. Lett. **127** (2021) 241602
Amplitudes and the Riemann Zeta Function Editors' Suggestion
arXiv:2108.07820
42. Achilleas P. Porfyriadis, **Grant N. Remmen** JHEP **10** (2021) 142
Horizon Acoustics of the GHS Black Hole and the Spectrum of AdS_2 arXiv:2106.10282
41. Ning Bao, Aidan Chatwin-Davies, **Grant N. Remmen** JHEP **7** (2021) 113
Entanglement Wedge Cross Section Inequalities from Replicated Geometries arXiv:2106.02640
40. Ning Bao, Jonathan Harper, **Grant N. Remmen** Phys. Rev. D **105** (2022) 026010
Holevo Information of Black Hole Mesostates arXiv:2103.06888
39. **Grant N. Remmen**, Nicholas L. Rodd Phys. Rev. D **105** (2022) 036006
Signs, Spin, SMEFT: Sum Rules at Dimension Six arXiv:2010.04723
38. Rafael Aoude et al. (including **Grant N. Remmen**) Snowmass 2021 Letter of Interest
On-Shell Methods for the SMEFT
37. Ning Bao, Aidan Chatwin-Davies, **Grant N. Remmen** JHEP **9** (2020) 102
Warping Wormholes with Dust: a Metric Construction of the Python's Lunch arXiv:2006.10762
36. **Grant N. Remmen**, Nicholas L. Rodd Phys. Rev. Lett. **125** (2020) 081601
Flavor Constraints from Unitarity and Analyticity arXiv:2004.02885
35. Clifford Cheung, **Grant N. Remmen** JHEP **5** (2020) 100
Entanglement and the Double Copy arXiv:2002.10470
34. Ning Bao, Aidan Chatwin-Davies, Jason Pollack, **Grant N. Remmen** JHEP **8** (2020) 65
Cosmological Decoherence from Thermal Gravitons arXiv:1911.10207
33. **Grant N. Remmen**, Nicholas L. Rodd JHEP **12** (2019) 32
Consistency of the Standard Model Effective Field Theory arXiv:1908.09845
32. Ning Bao, Aidan Chatwin-Davies, Jason Pollack, **Grant N. Remmen** JHEP **7** (2019) 152
Towards a Bit Threads Derivation of Holographic Entanglement of Purification arXiv:1905.04317
31. Clifford Cheung, Junyu Liu, **Grant N. Remmen** Phys. Rev. D **100** (2019) 046003
Entropy Bounds on Effective Field Theory from Rotating Dyon Black Holes arXiv:1903.09156
30. Raphael Bousso, Yasunori Nomura, **Grant N. Remmen** Phys. Rev. D **99** (2019) 046002
Outer Entropy and Quasilocal Energy arXiv:1812.06987
29. Ning Bao, Aidan Chatwin-Davies, **Grant N. Remmen** JHEP **2** (2019) 110
Entanglement of Purification and Multiboundary Wormhole Geometries arXiv:1811.01983

- PUBLICATIONS, 28. **Grant N. Remmen** Phys. Rev. D **98** (2018) 124008
CONTINUED *New Spacetimes for Rotating Dust in* arXiv:1810.12305
(2 + 1)-Dimensional General Relativity
27. Ning Bao, Aidan Chatwin-Davies, Jason Pollack, **Grant N. Remmen** JHEP **11** (2018) 71
Traversable Wormholes as Quantum Channels: arXiv:1808.05963
Exploring CFT Entanglement Structure and Channel Capacity in Holography
26. Yasunori Nomura, **Grant N. Remmen** JHEP **8** (2018) 63
Area Law Unification and the Holographic Event Horizon arXiv:1805.09339
25. Venkatesa Chandrasekaran, **Grant N. Remmen**, JHEP **11** (2018) 15
Arvin Shahbazi-Moghaddam arXiv:1804.03153
Higher-Point Positivity
24. Clifford Cheung, Junyu Liu, **Grant N. Remmen** JHEP **10** (2018) 4
Proof of the Weak Gravity Conjecture from Black Hole Entropy arXiv:1801.08546
23. Ning Bao, Sean M. Carroll, Aidan Chatwin-Davies, Phys. Rev. D **97** (2018) 126014
Jason Pollack, **Grant N. Remmen** arXiv:1712.04955
Branches of the Black Hole Wave Function Need Not Contain Firewalls
22. Chris Akers, Raphael Bousso, Illan F. Halpern, Phys. Rev. D **97** (2018) 024018
Grant N. Remmen arXiv:1711.06689
Boundary of the Future of a Surface
21. Clifford Cheung, **Grant N. Remmen**, Chia-Hsien Shen, Congkao Wen JHEP **4** (2018) 129
Pions as Gluons in Higher Dimensions arXiv:1709.04932
20. Clifford Cheung, **Grant N. Remmen** JHEP **9** (2017) 2
Hidden Simplicity of the Gravity Action arXiv:1705.00626
19. Sean M. Carroll, **Grant N. Remmen** Phys. Rev. D **95** (2017) 123504
A Nonlocal Approach to the Cosmological Constant Problem arXiv:1703.09715
18. Ning Bao, **Grant N. Remmen** EPL **121** (2018) 60007, Editor's Choice
Bulk Connectedness and Boundary Entanglement arXiv:1703.00018
17. Clifford Cheung, **Grant N. Remmen** JHEP **1** (2017) 104
Twofold Symmetries of the Pure Gravity Action arXiv:1612.03927
16. Clifford Cheung, **Grant N. Remmen** Phys. Rev. Lett. **118** (2017) 051601
Positivity of Curvature-Squared Corrections in Gravity arXiv:1608.02942
15. **Grant N. Remmen**, Ning Bao, Jason Pollack JHEP **7** (2016) 48
Entanglement Conservation, $ER = EPR$, arXiv:1604.08217
and a New Classical Area Theorem for Wormholes
14. Sean M. Carroll, **Grant N. Remmen** Phys. Rev. D **93** (2016) 124052
What is the Entropy in Entropic Gravity? arXiv:1601.07558
13. Clifford Cheung, **Grant N. Remmen** JHEP **4** (2016) 2
Positive Signs in Massive Gravity arXiv:1601.04068
12. Ning Bao, Jason Pollack, **Grant N. Remmen** JHEP **11** (2015) 126
Wormhole and Entanglement (Non-)Detection arXiv:1509.05426
in the $ER = EPR$ Correspondence
11. Brando Bellazzini, Clifford Cheung, **Grant N. Remmen** Phys. Rev. D **93** (2016) 064076
Quantum Gravity Constraints from Unitarity and Analyticity arXiv:1509.00851
10. Ning Bao, Jason Pollack, **Grant N. Remmen** Fortschr. Phys. **63** (2015) 705
Splitting Spacetime and Cloning Qubits: arXiv:1506.08203
Linking No-Go Theorems across the $ER = EPR$ Duality

PUBLICATIONS, CONTINUED	9. Ning Bao, ChunJun Cao, Sean M. Carroll, Aidan Chatwin-Davies, Nicholas Hunter-Jones, Jason Pollack, Grant N. Remmen <i>Consistency Conditions for an AdS Multiscale Entanglement Renormalization Ansatz Correspondence</i>	Phys. Rev. D 91 (2015) 125036 arXiv:1504.06632
	8. Clifford Cheung, Grant N. Remmen <i>Infrared Consistency and the Weak Gravity Conjecture</i>	JHEP 12 (2014) 87 arXiv:1407.7865
	7. Grant N. Remmen , Sean M. Carroll <i>How Many e-Folds Should We Expect from High-Scale Inflation?</i>	Phys. Rev. D 90 (2014) 063517 arXiv:1405.5538
	6. Clifford Cheung, Grant N. Remmen <i>Naturalness and the Weak Gravity Conjecture</i>	Phys. Rev. Lett. 113 (2014) 051601 arXiv:1402.2287
	5. Grant N. Remmen , Sean M. Carroll <i>Attractor Solutions in Scalar-Field Cosmology</i>	Phys. Rev. D 88 (2013) 083518 arXiv:1309.2611
	4. Grant N. Remmen , Kris Davidson, Andrea Mehner <i>Unexpected Ionization Structure in Eta Carinae’s “Weigelt Knots”</i>	Astrophys. J. 773 (2013) 27 arXiv:1302.2659
	3. Grant N. Remmen , Kinwah Wu <i>Complex Orbital Dynamics of a Double Neutron Star System Revolving around a Massive Black Hole</i>	Mon. Not. R. Astron. Soc. 430 (2013) 1940 arXiv:1301.2836
	2. Grant Remmen , Elwood McCreary <i>Measurement of the Speed and Energy Distribution of Cosmic Ray Muons</i>	JURP 25 (2012)
	1. Grant Remmen <i>A New Assessment of Dark Matter in the Milky Way Galaxy</i>	JURP 23 (2010)
TALKS	Amplitudes 2026 Queen Mary University of London	July 2026
	Stony Brook University, Simons Center for Geometry and Physics Physics Seminar	May 2026
	Tufts University Joint Tufts/MIT Cosmology Seminar	March 2026
	Online Swampland Seminar Series (IFT Madrid, CERN, Hamburg, Harvard) Swampland Seminar (virtual)	March 2026
	Scuola Normale Superiore, Pisa, Italy Theory Seminar (virtual)	February 2026
	UNC Chapel Hill High Energy Theory Seminar	February 2026
	UNC Chapel Hill Physics & Astronomy Department Colloquium	February 2026
	Niels Bohr Institute, University of Copenhagen Quantum Gravity Seminar (virtual)	January 2026
	Anhui University of Science & Technology, Center for Fundamental Physics, Huainan, China ASU-AUST-USTC Theoretical Physics Colloquium (virtual)	January 2026
	Ludwig-Maximilians-Universität, Munich Fields and Strings Seminar (virtual)	December 2025
	Carnegie Mellon University High Energy Physics Theory Seminar	November 2025
	Southeastern Regional Mathematical String Theory Meeting Virginia Tech	November 2025
	City College, CUNY High Energy Seminar	October 2025
	Stanford University LITP Colloquium	October 2025
	New York University, CCPP Brown Bag Seminar	September 2025
	University of Rhode Island Physics Colloquium	September 2025
	Lawrence Berkeley National Laboratory Particle Seminar	September 2025
	New York University, CCPP High Energy Physics Seminar	September 2025
	CERN TH Institute on Positivity, Amplitudes, and Phenomenology CERN	April 2025

TALKS, CONTINUED	Yale University High Energy Particle Theory Seminar	April 2025
	Institute for Advanced Study Amplitudes Group Meeting	March 2025
	Ohio State University Physics Department Colloquium	February 2025
	What is Particle Theory? Program Kavli Institute for Theoretical Physics	February 2025
	UC Davis, QMAP Fields, Strings, Gravity Seminar (virtual)	February 2025
	Texas A&M University Physics Department Colloquium	January 2025
	Texas A&M University High Energy Physics Seminar	January 2025
	Durham University, UK Amplitudes and Correlators Seminar (virtual)	December 2024
	Northeastern University High Energy Theory Seminar (virtual)	November 2024
	CERN ATLAS Electroweak Working Group Meeting (virtual)	September 2024
	Solvay Workshop on Near-Extremal Black Holes International Solvay Institutes and ULB, Brussels, Belgium	September 2024
	Modern Trends in Gravity and Black Holes Workshop University of Crete, Greece	June 2024
	Harvard University Swampland Seminar	May 2024
	Surveying the Landscape Workshop University of Massachusetts Amherst, ACFI	April 2024
	Johns Hopkins University High Energy Physics Theory Seminar	April 2024
	Particle Theory Initiative, What is String Theory? Program Kavli Institute for Theoretical Physics	March 2024
	California Institute of Technology High Energy Theory Seminar	March 2024
	University of Wisconsin–Madison Theory Seminar	February 2024
	University of Pennsylvania High Energy Theory Seminar	January 2024
	Columbia University Theory Seminar	January 2024
	University of Washington Particle Theory Seminar	November 2023
	Crete Center for Theoretical Physics High Energy Seminar (virtual)	November 2023
	New York University, CCPP Brown Bag Seminar	October 2023
	Boston University High Energy Theory Seminar	September 2023
	Swamplandia Workshop Instituto de Física Teórica, UAM-CSIC, and Harvard University, Madrid, Spain	September 2023
	Amplitudes 2023 CERN	August 2023
	Strings 2023 Perimeter Institute for Theoretical Physics	July 2023
	Kavli Institute for Theoretical Physics Generalized Symmetries Reading Group	June 2023
	Quark Confinement 2023 University of Minnesota Simons Collaboration on Confinement and QCD Strings	May 2023
	Kavli Institute for Theoretical Physics Locals' Lunch Talk	April 2023
	CERN Standard Model Electroweak Group Meeting, ATLAS Collaboration (virtual)	April 2023
	McGill University High Energy Theory Group Meeting (virtual)	April 2023
	University of Chicago, Kadanoff Center for Theoretical Physics Particle Theory Seminar	April 2023
	Princeton University High Energy Theory Seminar	March 2023
	California Amplitudes Meeting UC San Diego	March 2023

TALKS, CONTINUED	University of Michigan, LCTP High Energy Theory Seminar (two parts)	March 2023
	Indiana University High-Energy Physics/Astrophysics Seminar	March 2023
	California Institute of Technology Amplitudes Group Meeting	February 2023
	Bootstrapping Quantum Gravity Program Kavli Institute for Theoretical Physics	February 2023
	Stony Brook University, Simons Center for Geometry and Physics Special Physics Seminar	February 2023
	UC Davis, QMAP Particles/Cosmology Seminar	January 2023
	Brown University High Energy Theory Seminar (virtual)	November 2022
	Number Theory and Physics Workshop Simons Center for Geometry and Physics, Stony Brook University (virtual)	October 2022
	Institute for Advanced Study Amplitudes Group Meeting	October 2022
	Simons Symposium on Amplitudes Meet Cosmology Scotland	May 2022
	UC Santa Barbara High Energy and Gravity Seminar	May 2022
	Possible and Impossible in Effective Field Theory: From the S-Matrix to the Swampland Workshop Institute for Advanced Study	May 2022
	Argonne National Laboratory High Energy Physics Theory Seminar (virtual)	April 2022
	Kavli IPMU, University of Tokyo Mathematics - String Theory Seminar (virtual)	April 2022
	California Amplitudes Meeting UC Davis	March 2022
	California Institute of Technology Amplitudes Group Meeting	February 2022
	UC Irvine Particle Physics Seminar	January 2022
	QCD Meets Gravity Workshop UCLA (virtual)	December 2021
	Kavli Institute for Theoretical Physics Locals' Event	November 2021
	UC Santa Barbara High Energy and Gravity Seminar (virtual)	November 2021
	Abdus Salam International Centre for Theoretical Physics, Trieste, Italy High Energy, Cosmology, and Astroparticle Physics Seminar (virtual)	November 2021
	Brandeis University Quantum/Gravity Seminar (virtual)	November 2021
	ETH Zürich QFT, Strings and Beyond Seminar (virtual)	October 2021
	Perimeter Institute Quantum Fields and Strings Seminar (virtual)	October 2021
	Hertz Foundation Innovation Hour (virtual)	June 2021
	California Amplitudes Meeting UCLA (virtual)	March 2021
	New York University Physics Department Colloquium (virtual)	March 2021
	New York University Physics Research Seminar (virtual)	February 2021
	University of Florida High Energy Physics Seminar (virtual)	January 2021
	University of Chicago Particle Theory Seminar (virtual)	January 2021
	Korea Institute for Advanced Study High Energy Physics Seminar (virtual) Seoul, South Korea	December 2020
	UC Santa Barbara High Energy and Gravity Seminar (virtual)	November 2020
	Yale University Particle Theory Seminar (virtual)	October 2020
	Brookhaven National Laboratory High Energy Theory Seminar (virtual)	April 2020

TALKS, CONTINUED	Kavli IPMU, Univ. Tokyo Astronomy-Cosmology-Particle Physics Seminar (virtual)	April 2020
	UC Davis, QMAP Fields, Strings, Gravity Seminar (virtual)	April 2020
	The String Swampland and Quantum Gravity Constraints on Effective Theories Program Kavli Institute for Theoretical Physics	March 2020
	Brandeis University High-Energy and Gravitational Theory Chalk Talk	January 2020
	Brandeis University Physics Department Colloquium	January 2020
	University of Michigan, LCTP High Energy Theory Seminar	November 2019
	From Scattering to Expansion Workshop Northwestern University	October 2019
	UC Santa Barbara Particle Physics Phenomenology Seminar	October 2019
	UC Santa Barbara High Energy and Gravity Seminar	October 2019
	Navigating the Swampland Workshop Instituto de Física Teórica, UAM-CSIC, Madrid, Spain	September 2019
	University of Washington AdS/CFT Group Meeting	May 2019
	University of Washington Particle Theory Seminar	May 2019
	University of Minnesota, FTPI High Energy Theory Seminar	April 2019
	Stanford University Stanford Institute for Theoretical Physics Colloquium	April 2019
	UC Berkeley 4D Seminar	April 2019
	California Institute of Technology High Energy Theory Seminar	February 2019
	UC Davis Joint Theory Seminar	January 2019
	Harvard University Black Hole Initiative Colloquium	November 2018
	Cornell University Particle Theory Seminar	October 2018
	Institute for Advanced Study High Energy Theory Seminar	October 2018
	Vistas over the Swampland Workshop Instituto de Física Teórica, UAM-CSIC, Madrid, Spain	September 2018
	King's College London Special Seminar, Theoretical Particle Physics & Cosmology	June 2018
	Gravity, Cosmology & Physics Beyond the Standard Model Conference LPNHE, UPMC, Paris, France	June 2018
	Sakurai Thesis Prize Talk American Physical Society April Meeting, Columbus, OH <i>Quantum Gravity Constraints for Effective Field Theories</i>	April 2018
	University of Illinois, Urbana-Champaign Mathematical and Theoretical Physics Seminar	April 2018
	McGill University High Energy Theory Group Seminar (virtual)	March 2018
	California Institute of Technology High Energy Theory Seminar	February 2018
	California Institute of Technology Quantum Spacetime Meeting	February 2018
	UC Berkeley String Seminar	February 2018
	Stanford University Stanford Institute for Theoretical Physics Seminar	January 2018
	SLAC National Accelerator Laboratory Elementary Particle Physics Theory Seminar	October 2017
	Institute for Advanced Study High Energy Theory Seminar	October 2017

TALKS, CONTINUED	Massachusetts Institute of Technology String/Gravity Theory Seminar	May 2017
	California Institute of Technology Theoretical Physics Research Group Meeting	April 2017
	California Institute of Technology Theoretical Physics Journal Club	April 2017
	UC Berkeley String Seminar	February 2017
	QCD Meets Gravity Workshop UCLA	December 2016
	California Institute of Technology Theoretical Physics Research Group Meeting	November 2016
	California Institute of Technology Theoretical Physics Research Group Meeting	October 2016
	Johns Hopkins University High Energy Theory/Cosmology Seminar	October 2016
	California Institute of Technology Theoretical Physics Research Group Meeting	May 2016
	New York University High Energy Seminar	April 2016
	Harvard University Particle Theory Seminar	April 2016
	California Institute of Technology Theoretical Physics Research Group Meeting	February 2016
	California Institute of Technology Theoretical Physics Research Group Meeting	November 2015
	California Institute of Technology Theoretical Physics Journal Club	October 2015
	California Institute of Technology Theoretical Physics Research Group Meeting	April 2015
	California Institute of Technology Theoretical Physics Research Group Meeting	February 2015
	California Institute of Technology Theoretical Physics Research Group Meeting	October 2014
	California Institute of Technology Theoretical Physics Journal Club (two parts)	October 2014
	20 th International Symposium on Particles, Strings and Cosmology (PASCOS 2014) Warsaw, Poland	June 2014
	California Institute of Technology Theoretical Physics Research Group Meeting	May 2014
	California Institute of Technology Theoretical Physics Journal Club	February 2014
	California Institute of Technology Theoretical Physics Research Group Meeting	February 2014
	California Institute of Technology Theoretical Physics Journal Club	September 2013
	Hertz Foundation 50 th Anniversary Symposium Poster Presentation	August 2013
	American Physical Society April Meeting, Denver, CO	April 2013
	Mullard Space Science Laboratory, United Kingdom Theory Group Meeting	August 2011
	Dean's Summer International Student Day of Talks University College London, United Kingdom	August 2011
	217 th Meeting of the American Astronomical Society, Seattle, WA Poster Presentation	January 2011
CONFERENCE ORGANIZATION	Conference Co-Chair Kavli Institute for Theoretical Physics (virtual) <i>UV Meets the IR: Effective Field Theory Bounds from QFT to String Theory</i>	October 2020
SEMINAR ORGANIZATION	Organizer KITP Locals' Event Series	2022–2023
	Organizer UC Santa Barbara High Energy and Gravity Seminar Series	2020–2021
	Organizer UC Berkeley HEP-QIS Seminar Series	2018–2019
	Organizer UC Berkeley String Seminar Series	2017–2019

TEACHING	UC Santa Barbara, Department of Physics	Fall 2020
EXPERIENCE	Instructor and organizer of graduate short course <i>Impossible Physics: Constraining the Laws of Nature, from Scattering Amplitudes to Black Holes</i>	
	UC Berkeley, Department of Physics	April 2018
	Guest lecturer in Prof. Petr Hořava's Quantum Field Theory course	
	University of Minnesota, Institute of Technology, Department of Astronomy	Fall 2009
	Teaching assistant to Prof. Robert Gehrz, Department Chair	
JOURNAL	Physical Review Letters	
REFEREE	Physical Review D	
	Journal of High Energy Physics	
	Nuclear Physics B	
	Scientific Reports - Nature	
	Communications in Mathematical Physics	
	Annales Henri Poincaré	
THESES	Ph.D., Physics California Institute of Technology	Defended May 2017
	Grant Newton Remmen	
	<i>Defining Gravity: Effective Field Theory, Entanglement, and Cosmology</i>	
	Thesis advisors: Clifford Cheung and Sean M. Carroll, California Institute of Technology	
	B.S., Mathematics , <i>summa cum laude</i> University of Minnesota	Defended May 2012
	Grant N. Remmen	
	<i>Dynamics of a Rigid Spinning Ring in the Schwarzschild Metric:</i> <i>A Solution to a Gravitational Problem in Mathematical Physics</i>	
	Thesis advisor: Willard Miller, School of Mathematics, University of Minnesota	
	Research supervised by Kinwah Wu, Head of Theory, Mullard Space Science Laboratory, University College London.	
	B.S., Astrophysics , <i>summa cum laude</i> University of Minnesota	Defended December 2011
	Grant N. Remmen	
	Hubble Space Telescope <i>Subpixel Modeling of Anomalous</i> <i>High-Excitation Emission Lines in the Ejecta of Eta Carinae</i>	
	Thesis advisor: Kris Davidson, MN Institute for Astrophysics, University of Minnesota	
	B.S., Physics , <i>summa cum laude</i> University of Minnesota	Defended April 2010
	Grant Remmen	
	<i>Distortion of Black Holes caused by Motion relative to the Cosmic Microwave Background</i>	
	Thesis advisor: Robert Gehrz, Director, MN Institute for Astrophysics, University of Minnesota	
UNDERGRAD.	Hagstrum Award in Physics	2012
HONORS &	Outstanding Graduate in Mathematics	2012
AWARDS,	Franklin Scholarship in Physics	2011–2012
UNIVERSITY OF	Lando Scholarship in Mathematics	2011–2012
MINNESOTA	Richards Scholarship in Mathematics	2011–2012
	Nier Scholarship in Physics	2010–2011
	Thorp Scholarship in Mathematics	2010–2011
	Undergraduate Research Scholarship	2010
	Basford Award in Physics	2009–2010
	Institute of Technology Alumni Award	2009–2010
	Institute of Technology Honors Undergraduate Research Scholarship	2009
	Maroon & Gold Leadership Award	2008–2012
	3M/Alumni Award	2008–2012
	Bentson Scholar	2008–2012
	Dean's List , College of Science & Engineering/Institute of Technology	2008–2012
	McGraw Hill Student Achievement Recognition ,	2008
	Meritorious Work in General Chemistry	

SCIENCE	Outreach talk, <i>Quantum Field Theory is the Language of Theoretical Physics</i>	February 2025
OUTREACH	KITP Teachers' Conference: "This is Particle Theory"	
	Interacted with various major donors at the request of KITP	2022–2023
	Outreach talk for KITP administrative staff	May 2021
	Guest lecturer for Caltech's <i>Storytelling for Scientists</i> course	April 2021
	Presented talks on dark matter to physics classes in rural Minnesota	2011
TEST SCORES	GRE Physics—Perfect Score: 990/990	2011
	GRE General—Quantitative: 800/800, Verbal: 720/800, Analytical Writing: 5.5/6.0	2011
	SAT—Perfect Score: 2400/2400	2008
	SAT II—Perfect Scores: Math Level II 800/800 and Biology–Molecular 800/800	2008
LEADERSHIP, SERVICE, & CULTURAL ACTIVITIES	Co-author/-composer of a two-act musical, <i>From the Earth to the Moon</i> , based on the Verne novel	
	Caltech production of <i>From the Earth to the Moon</i>	
	Mainstage production, Assistant to the Director	2022
	Public reading (virtual), Music Director	2021
	California Institute of Technology Graduate Student Council Board of Directors	2013–2017
	Member, Academics Committee and Director at Large	2016–2017
	Member, Academics Committee and Physics Representative	2013–2016
	California Institute of Technology Faculty Library Committee, Student Representative	2013–2017
	Co-author/-composer of a two-act musical, <i>Boldly Go!</i> , a musical parody based upon <i>Star Trek</i>	
	Caltech production of <i>Boldly Go!</i>	
	Mainstage production, Music Director	2016
	Public reading, Music Director	2015
	University Study Abroad May Seminar: <i>Great Minds of the Renaissance</i> , Italy	2011
	History of Renaissance scientists (Galileo, da Vinci, etc.) and societal context	
	University of Minnesota Gospel Choir	2008–2010
	Detroit Lakes Community Summer Band Program	2008–2010
	University of Minnesota Honors Student Association	2008–2012
	University of Minnesota volunteer caller for Admissions Office	2008–2009
	U.S. Department of Education volunteer	2008
	Assembled hygiene kits for Washington, D.C. homeless	